pCloudy Banking App — API Documentation

Base URL: `https://demo-banking-server-production.up.railway.app`

Default Users

Username	Password	PIN
demo	password123	0000
user@test.com	Test@1234	1234
admin@test.com	Admin@1234	0000

> ***Note:*** *Protected APIs require* `Authorization: Bearer <token>` *header. Get the token from Login or Register response.*

---

1. Login

Method: `POST` **URL:** `https://demo-banking-server-production.up.railway.app/auth/login` **Auth Required:** No

Request Body

```
{
  "username": "demo",
  "password": "password123"
}
```

Success Response `200`

```
{
  "success": true,
  "token": "6f7c954b-6efd-419d-8c67-8a06f0f331b4",
  "user": {
    "name": "Demo User",
    "accountNo": "XX0000",
    "username": "demo"
  }
}
```

Failure Response `401`

```
{
  "success": false,
```

```
  "message": "Invalid credentials"
}
```

Postman

```
Method  : POST
URL     : https://demo-banking-server-production.up.railway.app/auth/login
Headers : Content-Type: application/json
Body    : raw → JSON
          { "username": "demo", "password": "password123" }
```

curl

```
curl -s -X POST https://demo-banking-server-production.up.railway.app/auth/login \
  -H "Content-Type: application/json" \
  -d '{"username":"demo","password":"password123"}'
```

---

2. Register

Method: `POST` **URL:** `https://demo-banking-server-production.up.railway.app/auth/register`
Auth Required: No

Request Body

```
{
  "username": "john",
  "password": "john@123",
  "name": "John"
}
```

Success Response `201`

```
{
  "success": true,
  "token": "abc123-...",
  "user": {
    "name": "John",
    "accountNo": "XX4823",
    "username": "john"
  }
}
```

Failure Response `409` — User already exists

```
{
  "success": false,
```

```
  "message": "Account already exists with this username"
}
```

Failure Response `400` — Missing fields

```
{
  "success": false,
  "message": "username, password and name are required"
}
```

Postman

```
Method  : POST
URL     : https://demo-banking-server-production.up.railway.app/auth/register
Headers : Content-Type: application/json
Body    : raw → JSON
          { "username": "john", "password": "john@123", "name": "John" }
```

curl

```
curl -s -X POST https://demo-banking-server-production.up.railway.app/auth/register
\
  -H "Content-Type: application/json" \
  -d '{"username":"john","password":"john@123","name":"John"}'
```

---

3. Logout

Method: `POST` **URL:** `https://demo-banking-server-production.up.railway.app/auth/logout`
Auth Required: Yes

Success Response `200`

```
{
  "success": true,
  "message": "Logged out successfully"
}
```

Postman

```
Method  : POST
URL     : https://demo-banking-server-production.up.railway.app/auth/logout
Headers : Authorization: Bearer <token>
Body    : none
```

curl

```
curl -s -X POST https://demo-banking-server-production.up.railway.app/auth/logout \
  -H "Authorization: Bearer YOUR_TOKEN"
```

---

4. Verify PIN

Method: `POST` **URL:** `https://demo-banking-server-production.up.railway.app/auth/verify-pin` **Auth Required:** Yes

Request Body

```
{
  "pin": "0000"
}
```

Success Response `200`

```
{
  "success": true,
  "message": "PIN verified"
}
```

Failure Response `401`

```
{
  "success": false,
  "message": "Invalid PIN"
}
```

Postman

```
Method  : POST
URL     : https://demo-banking-server-production.up.railway.app/auth/verify-pin
Headers : Authorization: Bearer <token>
          Content-Type: application/json
Body    : raw → JSON
          { "pin": "0000" }
```

curl

```
# Correct PIN
curl -s -X POST https://demo-banking-server-production.up.railway.app/auth/verify-
pin \
  -H "Content-Type: application/json" \
  -H "Authorization: Bearer YOUR_TOKEN" \
  -d '{"pin":"0000"}'
```

```
# Wrong PIN
curl -s -X POST https://demo-banking-server-production.up.railway.app/auth/verify-pin \
  -H "Content-Type: application/json" \
  -H "Authorization: Bearer YOUR_TOKEN" \
  -d '{"pin":"9999"}'
```

---

5. Get Balance

Method: `GET` **URL:** `https://demo-banking-server-production.up.railway.app/banking/balance`
Alias: `/account/balance` **Auth Required:** Yes

Success Response `200`

```
{
  "success": true,
  "balance": 24350,
  "currency": "INR",
  "accountNo": "XX0000",
  "accountHolder": "Demo User",
  "lastUpdated": "2026-03-18T10:00:00.000Z"
}
```

Postman

```
Method  : GET
URL     : https://demo-banking-server-production.up.railway.app/banking/balance
Headers : Authorization: Bearer <token>
Body    : none
```

curl

```
# Android path
curl -s https://demo-banking-server-production.up.railway.app/banking/balance \
  -H "Authorization: Bearer YOUR_TOKEN"

# Spec path (same response)
curl -s https://demo-banking-server-production.up.railway.app/account/balance \
  -H "Authorization: Bearer YOUR_TOKEN"
```

---

6. UPI Payment

Method: `POST` **URL:** `https://demo-banking-server-production.up.railway.app/banking/upi/pay` **Alias:** `/payment/upi` **Auth Required:** Yes

Request Body

```
{
  "upiId": "friend@upi",
  "amount": 100,
  "remarks": "Dinner"
}
```

Success Response `200`

```
{
  "success": true,
  "status": "SUCCESS",
  "message": "Payment successful",
  "transaction": {
    "id": "uuid",
    "type": "debit",
    "amount": -100,
    "description": "UPI Payment to friend@upi",
    "upiId": "friend@upi",
    "date": "2026-03-18T10:00:00.000Z",
    "status": "SUCCESS",
    "referenceId": "TXN1234567890"
  },
  "newBalance": 24250
}
```

> ***Auto Top-Up Behaviour*** *If the previous transaction caused the balance to fall below ₹5,000, the server automatically credits the account to bring the balance back to ₹10,000* ***before*** *processing the next payment.*
>
> - *An* `"Auto Balance Top-Up"` *credit transaction is inserted at the top of the transaction history.*
> - `newBalance` *in the response reflects the balance* ***after*** *the auto top-up and the current debit are both applied.*
>
> *Example:*
>
> - *Balance = ₹4,500 (below ₹5,000 after a previous payment)*
> - *Next payment of ₹200 → balance topped up to ₹10,000 first → deducted ₹200 →* `newBalance` *= ₹9,800*

Failure Response `422` — Inactive account

```
{
  "success": false,
  "message": "Payment failed. Beneficiary account is inactive.",
  "errorCode": "UPI_FAILURE_001"
}
```

Failure Response `422` — Insufficient balance

```
{
  "success": false,
  "message": "Insufficient balance",
  "errorCode": "INSUFFICIENT_FUNDS"
}
```

Test Cases

upiId	amount	Expected
`friend@upi`	`100`	✅ Success
`fail@test`	any	× 422 Inactive account
`timeout@test`	any	⏱ 408 Timeout (30s)
any	`999999`	× 422 Insufficient balance
any (when prev balance < ₹5,000)	any	✅ Success + auto top-up to ₹10,000 applied first

Postman

```
Method  : POST
URL     : https://demo-banking-server-production.up.railway.app/banking/upi/pay
Headers : Authorization: Bearer <token>
          Content-Type: application/json
Body    : raw → JSON
          { "upiId": "friend@upi", "amount": 100, "remarks": "Dinner" }
```

curl

```
# Success
curl -s -X POST https://demo-banking-server-
production.up.railway.app/banking/upi/pay \
  -H "Content-Type: application/json" \
  -H "Authorization: Bearer YOUR_TOKEN" \
  -d '{"upiId":"friend@upi","amount":100,"remarks":"Dinner"}'

# Failure — inactive account
curl -s -X POST https://demo-banking-server-
production.up.railway.app/banking/upi/pay \
  -H "Content-Type: application/json" \
  -H "Authorization: Bearer YOUR_TOKEN" \
  -d '{"upiId":"fail@test","amount":100}'

# Failure — insufficient balance
curl -s -X POST https://demo-banking-server-
production.up.railway.app/banking/upi/pay \
```

```
  -H "Content-Type: application/json" \
  -H "Authorization: Bearer YOUR_TOKEN" \
  -d '{"upiId":"friend@upi","amount":999999}'

# Spec path
curl -s -X POST https://demo-banking-server-production.up.railway.app/payment/upi \
  -H "Content-Type: application/json" \
  -H "Authorization: Bearer YOUR_TOKEN" \
  -d '{"upiId":"friend@upi","amount":100}'
```

---

7. Transactions

Method: `GET` **URL:** `https://demo-banking-server-production.up.railway.app/banking/transactions` **Alias:** `/transactions` **Auth Required:** Yes

Query Parameters

Param	Type	Default	Description
`page`	number	`1`	Page number
`limit`	number	`5`	Items per page
`type`	string	all	`debit` or `credit`

Success Response `200`

```
{
  "success": true,
  "transactions": [
    {
      "id": "uuid",
      "type": "debit",
      "amount": -100,
      "description": "UPI Payment to friend@upi",
      "upiId": "friend@upi",
      "date": "2026-03-18T10:00:00.000Z",
      "status": "SUCCESS"
    }
  ],
  "pagination": {
    "page": 1,
    "limit": 5,
    "total": 15,
    "totalPages": 3,
    "hasNext": true
  }
}
```

Postman

```
Method  : GET
URL     : https://demo-banking-server-production.up.railway.app/banking/transactions
Headers : Authorization: Bearer <token>
Params  : page=1, limit=5, type=debit   (optional)
Body    : none
```

curl

```
# Page 1 (default)
curl -s "https://demo-banking-server-production.up.railway.app/banking/transactions?page=1" \
  -H "Authorization: Bearer YOUR_TOKEN"

# Page 2
curl -s "https://demo-banking-server-production.up.railway.app/banking/transactions?page=2" \
  -H "Authorization: Bearer YOUR_TOKEN"

# Debits only
curl -s "https://demo-banking-server-production.up.railway.app/banking/transactions?type=debit" \
  -H "Authorization: Bearer YOUR_TOKEN"

# Credits only
curl -s "https://demo-banking-server-production.up.railway.app/banking/transactions?type=credit" \
  -H "Authorization: Bearer YOUR_TOKEN"

# Custom limit
curl -s "https://demo-banking-server-production.up.railway.app/banking/transactions?page=1&limit=10" \
  -H "Authorization: Bearer YOUR_TOKEN"

# Spec path
curl -s "https://demo-banking-server-production.up.railway.app/transactions?page=1" \
  -H "Authorization: Bearer YOUR_TOKEN"
```

---

8. Wallet Balance

Method: `GET` **URL:** `https://demo-banking-server-production.up.railway.app/wallet/balance`
Auth Required: Yes

Success Response `200`

```
{
  "success": true,
  "walletBalance": 5000,
  "currency": "INR"
}
```

Postman

```
Method  : GET
URL     : https://demo-banking-server-production.up.railway.app/wallet/balance
Headers : Authorization: Bearer <token>
Body    : none
```

curl

```
curl -s https://demo-banking-server-production.up.railway.app/wallet/balance \
  -H "Authorization: Bearer YOUR_TOKEN"
```

---

9. Wallet Topup

Method: `POST` **URL:** `https://demo-banking-server-production.up.railway.app/banking/topup`
Alias: `/wallet/topup` **Auth Required:** Yes

Request Body

```
{
  "amount": 500
}
```

Success Response `200`

```
{
  "success": true,
  "message": "Wallet topped up with ₹500",
  "walletBalance": 5500,
  "balance": 23850
}
```

Failure Response `422` — Insufficient bank balance

```
{
  "success": false,
  "message": "Insufficient bank balance for top-up"
}
```

Postman

```
Method  : POST
URL     : https://demo-banking-server-production.up.railway.app/banking/topup
Headers : Authorization: Bearer <token>
          Content-Type: application/json
Body    : raw → JSON
          { "amount": 500 }
```

curl

```
# Success
curl -s -X POST https://demo-banking-server-production.up.railway.app/banking/topup
\
  -H "Content-Type: application/json" \
  -H "Authorization: Bearer YOUR_TOKEN" \
  -d '{"amount":500}'

# Failure — insufficient balance
curl -s -X POST https://demo-banking-server-production.up.railway.app/banking/topup
\
  -H "Content-Type: application/json" \
  -H "Authorization: Bearer YOUR_TOKEN" \
  -d '{"amount":999999}'

# Spec path
curl -s -X POST https://demo-banking-server-production.up.railway.app/wallet/topup \
  -H "Content-Type: application/json" \
  -H "Authorization: Bearer YOUR_TOKEN" \
  -d '{"amount":1000}'
```

---

10. Notifications

Method: `GET` **URL:** `https://demo-banking-server-production.up.railway.app/banking/notifications` **Alias:** `/notifications` **Auth Required:** Yes

Success Response `200`

```
{
  "success": true,
  "notifications": [
    {
      "id": "uuid",
      "type": "LOW_BALANCE",
      "title": "Low Balance Alert",
      "body": "Your account balance ₹4350.00 is below ₹5,000",
      "deepLink": "banking://wallet/topup",
      "timestamp": "2026-03-18T10:00:00.000Z",
```

```
      "read": false
    },
    {
      "id": "uuid",
      "type": "INFO",
      "title": "Security Tip",
      "body": "Never share your PIN or OTP with anyone.",
      "deepLink": "banking://profile/security",
      "timestamp": "2026-03-18T09:00:00.000Z",
      "read": true
    }
  ],
  "unreadCount": 1
}
```

> ***Note:*** *`LOW_BALANCE` alert appears only when balance is below ₹5,000*

Postman

```
Method  : GET
URL     : https://demo-banking-server-production.up.railway.app/banking/notifications
Headers : Authorization: Bearer <token>
Body    : none
```

curl

```
# Android path
curl -s https://demo-banking-server-production.up.railway.app/banking/notifications
\
  -H "Authorization: Bearer YOUR_TOKEN"

# Spec path
curl -s https://demo-banking-server-production.up.railway.app/notifications \
  -H "Authorization: Bearer YOUR_TOKEN"
```

---

11. Profile

Method: `GET` **URL:** `https://demo-banking-server-production.up.railway.app/profile` **Auth Required:** Yes

Success Response `200`

```
{
  "success": true,
  "profile": {
    "name": "Demo User",
    "username": "demo",
    "accountNo": "XX0000",
```

```
    "linkedAccounts": [
      { "bank": "SBI",  "accountNo": "XXXX1234", "isPrimary": true },
      { "bank": "HDFC", "accountNo": "XXXX5678", "isPrimary": false }
    ]
  }
}
```

Postman

```
Method  : GET
URL     : https://demo-banking-server-production.up.railway.app/profile
Headers : Authorization: Bearer <token>
Body    : none
```

curl

```
curl -s https://demo-banking-server-production.up.railway.app/profile \
  -H "Authorization: Bearer YOUR_TOKEN"
```

Quick Test — Login + All APIs in one shot

```
# Step 1 — Login and save token
export TOKEN=$(curl -s -X POST \
  https://demo-banking-server-production.up.railway.app/auth/login \
  -H "Content-Type: application/json" \
  -d '{"username":"demo","password":"password123"}' \
  | grep -o '"token":"[^"]*"' | cut -d'"' -f4)
echo "Token: $TOKEN"

# Step 2 — Run all APIs
curl -s https://demo-banking-server-production.up.railway.app/banking/balance -H
"Authorization: Bearer $TOKEN"
curl -s "https://demo-banking-server-production.up.railway.app/banking/transactions?
page=1" -H "Authorization: Bearer $TOKEN"
curl -s https://demo-banking-server-production.up.railway.app/wallet/balance -H
"Authorization: Bearer $TOKEN"
curl -s https://demo-banking-server-production.up.railway.app/banking/notifications
-H "Authorization: Bearer $TOKEN"
curl -s https://demo-banking-server-production.up.railway.app/profile -H
"Authorization: Bearer $TOKEN"

# Step 3 — UPI Payment
curl -s -X POST https://demo-banking-server-
production.up.railway.app/banking/upi/pay \
  -H "Content-Type: application/json" \
  -H "Authorization: Bearer $TOKEN" \
  -d '{"upiId":"friend@upi","amount":100}'
```

```
# Step 4 — Wallet Topup
curl -s -X POST https://demo-banking-server-production.up.railway.app/banking/topup
\
  -H "Content-Type: application/json" \
  -H "Authorization: Bearer $TOKEN" \
  -d '{"amount":500}'

# Step 5 — Logout
curl -s -X POST https://demo-banking-server-production.up.railway.app/auth/logout \
  -H "Authorization: Bearer $TOKEN"

# Step 6 — Verify token invalid after logout (should return 401)
curl -s https://demo-banking-server-production.up.railway.app/banking/balance \
  -H "Authorization: Bearer $TOKEN"
```

Multi-Device Test

```
# Login 3 users (3 different devices)
export TOKEN1=$(curl -s -X POST https://demo-banking-server-
production.up.railway.app/auth/login \
  -H "Content-Type: application/json" \
  -d '{"username":"demo","password":"password123"}' \
  | grep -o '"token":"[^"]*"' | cut -d'"' -f4)

export TOKEN2=$(curl -s -X POST https://demo-banking-server-
production.up.railway.app/auth/login \
  -H "Content-Type: application/json" \
  -d '{"username":"user@test.com","password":"Test@1234"}' \
  | grep -o '"token":"[^"]*"' | cut -d'"' -f4)

export TOKEN3=$(curl -s -X POST https://demo-banking-server-
production.up.railway.app/auth/login \
  -H "Content-Type: application/json" \
  -d '{"username":"admin@test.com","password":"Admin@1234"}' \
  | grep -o '"token":"[^"]*"' | cut -d'"' -f4)

echo "Device1 (demo):           $TOKEN1"
echo "Device2 (user@test.com):  $TOKEN2"
echo "Device3 (admin@test.com): $TOKEN3"

# Check all 3 balances — each user has own balance
echo "--- Balances ---"
curl -s https://demo-banking-server-production.up.railway.app/banking/balance -H
"Authorization: Bearer $TOKEN1"
curl -s https://demo-banking-server-production.up.railway.app/banking/balance -H
"Authorization: Bearer $TOKEN2"
curl -s https://demo-banking-server-production.up.railway.app/banking/balance -H
"Authorization: Bearer $TOKEN3"
```

```
# Device1 pays — only Device1 balance drops
curl -s -X POST https://demo-banking-server-
production.up.railway.app/banking/upi/pay \
  -H "Content-Type: application/json" \
  -H "Authorization: Bearer $TOKEN1" \
  -d '{"upiId":"friend@upi","amount":5000}'

# Verify Device2 and Device3 are unaffected
echo "--- After Device1 payment ---"
curl -s https://demo-banking-server-production.up.railway.app/banking/balance -H
"Authorization: Bearer $TOKEN1"
curl -s https://demo-banking-server-production.up.railway.app/banking/balance -H
"Authorization: Bearer $TOKEN2"
curl -s https://demo-banking-server-production.up.railway.app/banking/balance -H
"Authorization: Bearer $TOKEN3"
```

HTTP Status Codes

Code	Meaning
`200`	Success
`201`	Created (register)
`400`	Bad request — missing fields
`401`	Unauthorized — wrong credentials or invalid/expired token
`408`	Timeout — `timeout@test` UPI ID
`409`	Conflict — username already exists
`422`	Payment failed / Insufficient balance

All Endpoints Summary

#	Method	Endpoint	Auth	Description
1	POST	`/auth/login`	No	Login
2	POST	`/auth/register`	No	Register new account
3	POST	`/auth/logout`	Yes	Logout
4	POST	`/auth/verify-pin`	Yes	Verify PIN
5	GET	`/banking/balance`	Yes	Get account balance
6	GET	`/account/balance`	Yes	Get balance (spec alias)
7	POST	`/banking/upi/pay`	Yes	UPI payment
8	POST	`/payment/upi`	Yes	UPI payment (spec alias)

9	GET	`/banking/transactions`	Yes	Transaction history
10	GET	`/transactions`	Yes	Transactions (spec alias)
11	GET	`/wallet/balance`	Yes	Wallet balance
12	POST	`/banking/topup`	Yes	Wallet topup
13	POST	`/wallet/topup`	Yes	Wallet topup (spec alias)
14	GET	`/banking/notifications`	Yes	Notifications
15	GET	`/notifications`	Yes	Notifications (spec alias)
16	GET	`/profile`	Yes	User profile